Design System Rebuild — From Scattered Files to Governed Infrastructure

Role: UI Designer, Design Systems | **Timeline:** 2023–2025 | **Scope:** 6 products, 11 teams, enterprise cybersecurity ($400M+ ARR)

30-Second Summary

An enterprise cybersecurity company selling to every US cabinet-level agency had a design system that was a shared file with no governance. Products looked different, broke in dark mode, and failed the accessibility audits required for federal procurement. I rebuilt it.

Detach rate: 40% → under 1%

Component usage: 7,000 → 61,000+ insertions per quarter

Token coverage: 791 variables, light and dark modes

Adoption: 2 products → 6 products across 11 teams

Compliance: AA+ contrast, Section 508 alignment

All numbers verified via design analytics tooling.

What Was Broken

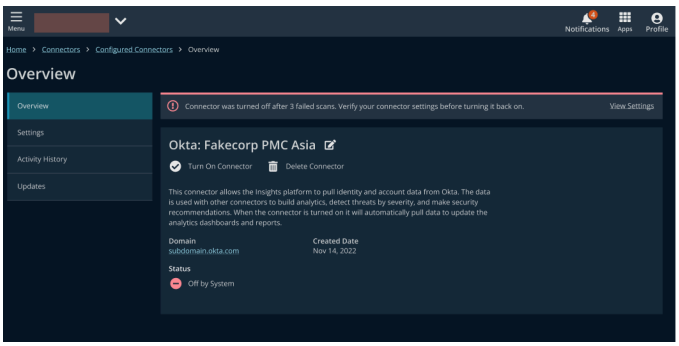
The design system was a collection of disconnected files. I audited the library and found 100+ components that had drifted from engineering implementation. No token layer existed. Dark mode was handled per-product with hardcoded values. Federal accessibility audits (required for Section 508 compliance and VPAT

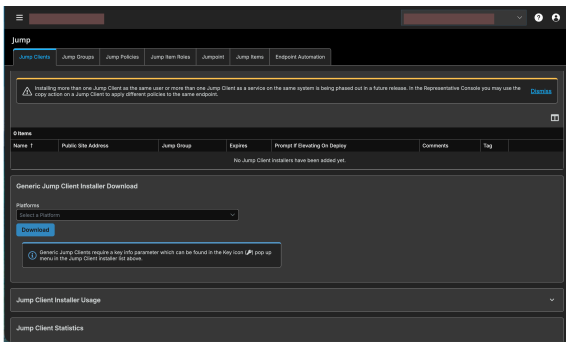
maintenance) produced inconsistent results because there was no shared standard.

Designers detached components to customize them at a 40% rate. That meant every product was slowly becoming its own system. Engineering built the same UI twice because the design source did not match production.

What I Owned

I owned the audit and component rebuild, the three-layer token architecture, accessibility implementation, documentation migration, and adoption measurement. I owned the governance model from day one and still do. That includes defining which tokens apply where, auditing usage across teams, maintaining the system post-release, and educating designers and engineers on how to build with it correctly. Post-release, I extended the system's role into AI-assisted workflows, adding another layer of structure to how the system gets used.





Decisions I Made

Three-layer token architecture (Primitive → Semantic → Component). Six products running on different frontend frameworks needed an abstraction layer between raw values and component-level usage. I chose a naming convention that gave engineering unambiguous references: `{category}.{role}.{state}.{mode}`.

Governance model with a formal entry gate. Without it, teams duplicated components under different names. A "Tiles vs Cards" conflict was the clearest example. I resolved it by defining Tiles as navigation triggers and Cards as content containers, which also clarified accessibility focus state requirements.

Documentation migration. The previous documentation tool treated docs as a separate artifact from code. I moved usage guidance, token references, and code demos into a unified framework so designers and engineers read the same source of truth. Documentation became versioned alongside the component code.

Constraints

Legacy frameworks: two products ran older UI kits that could not consume tokens natively. I created migration guides mapping hardcoded values to semantic tokens, making adoption low-effort.

No dedicated frontend engineer: I scoped component specs to minimize ambiguity. Figma component properties mapped 1:1 to code props.

Federal compliance pressure: the company maintains 16 VPATs for government procurement. Accessibility failures can disqualify the company from contracts worth an estimated $60M–$140M (based on public revenue disclosures).

What Shipped

Visual modernization with AA+ contrast across light and dark modes. Version-controlled releases with change logs. Component lifecycle management. Analytics dashboards tracking insertions, detaches, and usage patterns per team. Adoption across 11 teams. 90% system-wide adoption. 94,000+ tokens in production.

What I Still Question

Adoption metrics confirm teams use the system. They do not confirm teams understand it well enough to extend it correctly without me. The gap between usage and genuine shared ownership is unsolved.

Why This Mattered to the Business

Revenue protection. The company sells to US federal agencies under contracts that require Section 508 compliance and VPAT maintenance. Stale accessibility documentation and inconsistent UI across products create disqualification risk during procurement reviews. The estimated contract exposure is $60M–$140M

based on public revenue disclosures. A governed, accessible design system is not a design preference. It is procurement infrastructure.

Engineering time recovered. Before the rebuild, designers detached components at a 40% rate. Every detach means engineering builds custom UI instead of using the system. At scale across 11 teams and 6 products, that is hundreds of hours per quarter spent rebuilding what should already exist. Reducing detach to under 1% returned that time to feature work.

Faster product delivery. New product surfaces that previously required weeks of custom component work now ship using system components. The documentation migration means engineers spend less time asking designers "how does this work?" and more time building. Design reviews shortened because the system enforces consistency before the review happens.

Reduced onboarding cost. New designers inherit a governed system with versioned docs instead of tribal knowledge. Ramp-up shortened across all 11 teams.

This case study connects to the Inline Alert (feedback taxonomy, cross-team adoption) and Token Architecture (three-layer model, naming conventions, export pipeline) work documented separately.